

Lesson 4-3

Methods for Evaluating Websites

Learning Objectives

- Distinguish qualitative from quantitative evaluation, and explain the major indicators used in web analytics.
- Explain how heuristic evaluation works, and why combining both approaches gives a clearer picture of a website.

How do you know whether a website works well for its users? You can watch real users try it and ask what confuses them — that is qualitative evaluation. Or you can count how many people visit, how long they stay, and how many buy — that is quantitative evaluation. Numbers tell you how much is happening; users' voices tell you why. The best teams use both, together with expert checks, to see a site clearly and improve it.

? Guiding question: How can a website be evaluated, and why is it useful to combine numbers with users' voices?

Key Fact:
Websites are evaluated qualitatively (why — usability tests, interviews, heuristic checks) and quantitatively (how much — web analytics: PV, bounce rate, CVR); combining both gives the clearest picture.

Key Concepts:

- qualitative evaluation
- quantitative evaluation
- web analytics
- PV / bounce rate / CVR
- heuristic evaluation

KEY TERMS
qualitative evaluation — the "why" from users' voices and behavior.
quantitative evaluation — the "how much" from numerical data.



Qualitative watches users; quantitative reads numbers.

Point!

1. Qualitative and Quantitative Evaluation

(1) Methods for evaluation generally fall into two main categories: **(qualitative evaluation)** and **(quantitative evaluation)**. These evaluation methods are used in a wide range of applications, and the same concepts apply to website evaluation.

Evaluation method	Description (purpose)	Specific techniques
Qualitative evaluation	Explore in depth the "reasons" behind users' psychology and behavior, and evaluate qualitative aspects that cannot be expressed in numbers.	• Usability testing: observe actual operations to identify points that are hard to use. • Interviews: hear users' direct voices to understand their needs and dissatisfactions.
Quantitative evaluation	Use numerical data such as access counts and time, and evaluate the current state objectively and statistically.	• Web analytics: analyze metrics such as PV count and time spent. • A/B testing: compare two options to verify which leads to better results.

(2) Qualitative evaluation is suited to exploring the cause of "why is it hard to use," while quantitative evaluation is suited to grasping the overall trend of "how much is it being used." By combining both, more accurate evaluation is possible.

PAUSE & THINK

Would counting visits tell you *why* people leave a page? Which method would?

2. Major Indicators in Web Analytics

(1) **(Web analytics)**…a technique for analyzing how a website is accessed using numerical values; it is the central method of quantitative evaluation for websites. The following indicators are commonly used.

Indicator	Meaning	Description
(PV (Page View))	The total number of times a page has been viewed.	A basic indicator that measures how widely the site is seen and how much exposure it gets.
(Bounce rate)	The proportion of visits that ended after viewing only the first page.	An indicator of whether the page content matched the user's expectations and whether further actions were prompted.
(Conversion rate (CVR))	The proportion of visits that achieved a target action (purchase, registration, etc.).	A very important indicator that measures how much the site contributes to business results (profits).

3. Heuristic Evaluation

(1) **(Heuristic evaluation)**…a method in which experts identify problems in a website's UI (user interface) based on rules of thumb (heuristics). It is a form of qualitative evaluation, used along with usability testing and other methods to investigate the causes of usability problems.

(2) Major evaluation perspectives

① Consistency…whether the design and operations are unified across the site.

② Error prevention…whether the design makes it hard for users to make mistakes.

③ Visibility of system status…whether system status is clearly indicated to users (e.g., processing indicators, input error messages).

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Qualitative evaluation is a method that uses numerical data to evaluate quantitatively.

B PV (Page View) is an indicator showing the number of times a page has been viewed.

C The bounce rate is an indicator showing the proportion of visits that achieved a target action.

D Heuristic evaluation is a method in which experts identify UI problems based on rules of thumb.

(2) Match each description (a–c) with the most appropriate indicator from the choices below (A–C).

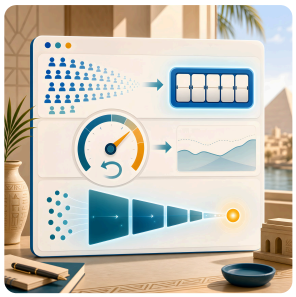
a The number of times a page has been viewed

b The proportion of visits that ended after viewing only the first page

c The proportion of visits that achieved a target action

[Choices] **A** PV (Page View) **B** Bounce rate **C** Conversion rate

✓ Solution



Analytics tracks visits, bounce, and conversions.

KEY TERMS

PV — total page views.
bounce rate — the share of visits that end after only one page.
CVR — the share of visits that reach the target action.



Heuristic review checks a site against usability rules.

PAUSE & THINK

Match the question to the method: “how many?” vs “why?”

(1)

A Qualitative evaluation observes user voices and behavior to evaluate qualitatively. Using numerical data is quantitative evaluation. Therefore, ×.

B Correct definition of PV. Therefore, ○.

C The bounce rate is the proportion of visits that ended after viewing only the first page. The proportion of target action achievement is the conversion rate. Therefore, ×.

D Correct definition of heuristic evaluation. Therefore, ○.

(2)

a: The number of page views is PV. **A**

b: The proportion who left after only the first page is the bounce rate. **B**

c: The proportion who achieved a target is the conversion rate. **C**

Try

1 Answer the following questions.

1. What is the term for the method that evaluates qualitatively by observing user voices and behavior?
2. What is the term for the method that evaluates quantitatively using numerical data?
3. What is the two-letter abbreviation for the indicator showing the number of times a page has been viewed?
4. What is the term for the method in which experts identify UI problems based on rules of thumb?

2 Answer the following questions.

(1) From the following options (A–D), choose all terms that are part of quantitative evaluation methods.

A Usability testing

B Web analytics

C Interviews

D A/B testing

EXAM-STYLE QUESTION

Explain why a website team should use both qualitative and quantitative evaluation rather than only one. **[6] Refer to:** what numbers show and what users' voices show.

PAUSE & THINK

Which of the four above rely on numbers, and which on people's words?

(2) From the following options (A–D), choose the one that most appropriately describes the conversion rate.

A The proportion of times a page has been viewed

B The proportion of visits that ended after viewing only the first page

C The proportion of visits that achieved a target action (product purchase, member registration, etc.)

D The proportion of viewed pages relative to the total number of pages on the site

Think as an Engineer — Investigate & Decide

A small online bookstore in your city wants to know whether its website really works for customers. You plan how to evaluate it.

☐ **Rely on the owner's opinion** ☐ **Use both quantitative and qualitative methods**

1 · Collect data. Ask five classmates who have used an online bookstore to visit any online shop and, for each, record one number you could measure (e.g., how many pages they viewed) and one thing they said was confusing. Put the results in a small table.

2 · Analyze the signals. Suppose the bookstore's PV is high but its conversion rate is very low. What might that mean — and which method (quantitative or qualitative) would tell you *why*? Then consider one customer the data may hide — for example, someone shopping only on a small phone screen, or a first-time visitor — and note one difficulty they might face that a raw PV count would never show.

3 · Decide. Recommend which two evaluation methods the bookstore should use first and give two reasons drawn from the table you built in step 1. **In pairs,** compare your choices and agree together on the single most useful indicator to track.

Give evidence for your answer.

Stuck? Start from the gap in the data: a high PV with a low CVR means people arrive but don't act — so the numbers only point to *where* to look. Pick the qualitative method (a short usability test or an interview) that answers the exact question your numbers raised.

🌍 In a New Context

A school's new website gets many visits, but most people leave after the first page. Name one quantitative indicator that reveals this problem, one qualitative method to find its cause, and one likely fix.

❓ Lesson Question — Answered

The lesson asked how a website can be evaluated and why numbers and users' voices are combined. Evaluation falls into two kinds. Quantitative evaluation uses numerical data — through web analytics indicators such as PV (how many views), bounce rate (how many leave after the first page), and CVR (how many reach a target action) — to show how much a site is used. Qualitative evaluation explores the reasons behind users' behavior through usability testing, interviews, and heuristic evaluation, in which experts check the UI against rules of thumb. Numbers reveal what is happening but not why; users' voices and expert checks reveal the causes. Combining both — the "how much" and the "why" — gives the most accurate evaluation and the clearest path to improvement.

KEY TERMS

web analytics — numbers on how a site is accessed.
heuristic evaluation — experts check the UI by rules of thumb.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Methods for evaluating a website include (a) evaluation, which observes user voices and behavior, and (b) evaluation, which analyzes using numerical data. Typical indicators used in (b) evaluation are (c), which shows the number of times a page has been viewed; (d), which shows the proportion of visits that ended after viewing only the first page; and (e), which shows the proportion of visits that achieved a target action.

1. Fill in the blanks (a)–(e).

3 Answer the following questions.

(1) For each of evaluation methods (a–c), indicate which it falls under: **A** Qualitative evaluation, **B** Quantitative evaluation.

a Have users actually explore the website and observe its usability

b Tally the access counts and page views of the website

c Interview users to hear their impressions and improvement requests

⭐ REFLECT & CHALLENGE

Reflect: If you could see only one metric for a site you use, would you choose PV, bounce rate, or CVR — and why?
Challenge: A page has a high bounce rate. Design a short qualitative test that would reveal the cause, and say what you would change first.

(2) From the following options (A–D), choose the one that most appropriately describes heuristic evaluation.

A A method in which users explore the site and usability is observed

B A method in which numerical data is tallied and site usage is analyzed

C A method in which experts identify UI problems based on rules of thumb

D A method in which two patterns are prepared and compared with data to see which is more effective

Key takeaway

Remember:
Quantitative (PV, bounce rate, CVR) shows how much; qualitative (usability tests, interviews, heuristic checks) shows why. Combine both to evaluate a site clearly.